

Population Change and Public Transport Service Provision in Metropolitan Victoria

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Section 1: Introduction

The narrative visualisation aims to show the relationship between population change and public transport service provision across Statistical Area Level 2 (SA2) regions in metropolitan Victoria. The primary objective of the visualisations is to explore whether areas which experienced strong population growth historically also receive comparatively higher levels of public transport service provision and to identify potential mismatches between population growth and transport accessibility.

The visualisations intend to communicate the three key messages from the analysis such that population growth is unevenly distributed across metropolitan Victoria, with higher growth concentrated in outer metropolitan areas, public transport service intensity varies substantially between SA2s and does not always align with population change patterns, and some historically high-growth areas may experience comparatively lower levels of transport service provision, highlighting regions where infrastructure and service supply may be struggling to keep pace with population demand.

The intended audience is the general public with an interest in urban planning, transport, and population dynamics, including policymakers, transport planners, students, and residents of metropolitan Victoria. The audience is assumed to have limited technical knowledge of spatial or transport datasets but possesses a general understanding of maps, charts, and geographic regions. Consequently, the visualisation prioritises clarity, intuitive interaction, and contextual explanations over statistical complexity.

To support exploration, the dashboard integrates multiple linked visualisations, including a choropleth map, scatterplot, time-series, ranking, and detailed information panel. Interactive filtering and linked highlighting allow users to investigate both metropolitan wide patterns and individual SA2 regions, enabling comparison between population change and transport service provision at multiple spatial and temporal scales.

Section 2: Design Process

The design process involved the development and evaluation of five design sheets, each exploring different combinations of visualisations based on narrative goals, audience needs, and principles from visual analytics. The target audience consisted of members of the general public, students, and policymakers interested in understanding how effectively public transport services support high-growth regions across metropolitan Victoria. As this audience was assumed to possess limited technical expertise, the design prioritised accessibility, intuitive interactions, and clear visual encodings.

Throughout the design process, the main objective to communicate the relationship between population change and public transport service provision across SA2s in metropolitan Victoria remained consistent. Different narrative structures were explored to determine which approach best balanced concepts of storytelling, user control, and analytical depth for the intended audience.

Sheet 1 (Brainstorm):

The first design sheet explored several possible visualisation approaches for communicating the relationship between population change and service provision. Early concepts included maps, bar charts, scatterplots, line charts, and other exploratory visualisations.

The main challenge identified during this stage was balancing geographic context with quantitative comparison. Following Munzner's What-Why-How framework (Munzner, 2014), the primary task was identified as enabling comparison and exploration of relationships between spatial, temporal, demographic, and transport variables. This sheet established the importance of preserving both geographic and analytical perspectives throughout the narrative.

Sheet 2 (Initial Design 1):

The second design sheet adopted a scrollytelling narrative combining a choropleth map and heat map. This design approach was intended to guide users through a predefined sequence of insights while progressively revealing information.

The choropleth map was used to display spatial patterns of population change over time, while the heat map was intended to highlight variation in service provision across regions. Together, these visualisations offered strong narrative control and spatial pattern recognition while supporting the communication of key findings in a structured narrative sequence. The design provided strong authorial control and effectively introduced the broader spatial context of the dataset.

However, although the scrollytelling format effectively guided users through the story, it limited opportunities for exploration. Users could observe patterns but had relatively little control over investigating individual SA2s or comparing regions of personal interest. Additionally, the maps effectively conveyed spatial context, but there was no explicit quantification of how well regions were served. Exact comparisons were therefore difficult, particularly when differences were small. These limitations suggested a need for greater analytical depth and user-driven exploration.

Sheet 3 (Initial Design 2):

The third design sheet shifted the narrative towards a drill-down structure, combining a scatterplot, time-series chart, diverging bar chart, and detail panel.

The scatterplot was introduced as the main visualisation for examining the relationship between population change and service intensity. Position along a common scale is one of the most accurate visual encodings (Cleveland & McGill, 1984), making the scatterplot particularly effective and appropriate for identifying trends and outliers between the two variables.

The time-series chart provided temporal context by allowing users to investigate population trends within individual SA2 regions. The diverging bar chart was designed to compare observed service intensity against a derived expected service intensity benchmark, supporting interpretation of whether public transport service appeared sufficient relative to population growth.

The design substantially increased analytical depth and user control. However, the absence of a geographic view reduced the effectiveness of the narrative. While users could perform detailed exploration, they lacked the spatial context necessary to understand broader metropolitan patterns. Furthermore, the construction of an expected service intensity metric

may prove difficult to justify and communicate to non-technical audiences. These issues highlighted the need for a design that combined analytical depth with geographic context while also considering the technical level of the audience.

Sheet 4 (Initial Design 3):

The fourth design sheet explored an interactive slideshow format featuring a histogram, box plot, and ranked dot plot.

The goal was to guide users through a sequence of analytical perspectives while maintaining some degree of interaction. The histogram and boxplot focused on distributional characteristics of service intensity across groups of regions categorised by population change, reducing visual clutter while emphasising overall trends. The ranked dot plot supported direct comparison between individual regions and provided a clearer understanding of relative service provision.

This approach improved the communication of statistical summaries and distributions. However, the slideshow structure fragmented the analysis by separating related views across multiple slides. Consequently, users could not easily compare geographic, temporal, and statistical perspectives simultaneously, increasing cognitive effort required to synthesise findings. Similar to design sheet 3, the lack of a strong spatial component weakened the overall effectiveness of the narrative.

Sheet 5 (Final Design):

The final design integrated the strongest elements from the previous sheets into a coordinated multi-view dashboard with a drill-down narrative structure. The final design combined a choropleth map for spatial overview, a scatterplot for analysing relationships between variables, a time-series chart for temporal exploration, a ranked dot plot for benchmarking service intensity, and a detail panel providing contextual interpretation.

This design reflects Shneiderman's visual information-seeking mantra of "overview first, zoom and filter, then details-on-demand" (Shneiderman, 1996). Users begin with a metropolitan overview through the map and scatterplot, with the ability to filter by SA4 regions, and then investigate individual SA2s through linked views and the detail panel.

Following the What-Why-How framework (Munzner, 2014), the visualisations were designed around several key data attributes including population change, service intensity, geography, and time. The primary analytical tasks were comparison, exploration, identification of outliers, and investigation of potential mismatches between population growth and service provision. These tasks informed the selection of the choropleth map, scatterplot, time-series chart, and ranked dot plot, which together provide complementary views of the same underlying relationships.

Several interaction techniques were implemented to support this workflow. Linked highlighting ensures that selections made on the map or scatterplot propagate across all visualisations, reinforcing connections between views. Filtering allows users to focus on specific SA4 regions, reducing visual clutter and supporting focused exploration, while the detail panel provides additional interpretation without overwhelming the main visualisations.

Design Justification:

The final dashboard was selected because it provides the strongest balance between author-driven storytelling and reader-driven exploration. Unlike the scrollytelling and interactive slideshow designs, users are not constrained to a predetermined sequence of views. Instead, they can investigate regions of personal interest while still being guided by the overall narrative structure and supporting exploratory text. This flexibility was considered important given the diverse needs of the intended audience.

The final design adopted coordinated multiple views, where several linked visualisations provide complementary perspectives on the same data. The choropleth map provides a spatial overview of population change, the scatterplot supports analysis of the relationship between population growth and transport service intensity, the time-series chart provides temporal context, and the ranked dot plot enables comparisons of service intensity between metropolitan Victoria SA2s. Linked interactions ensure that selections made in one view are reflected across all other views, supporting efficient exploration and reducing the cognitive effort required to connect information across visualisations.

Gestalt principles were applied throughout the layout design. The principles of proximity and enclosure were used to group related visualisations into clearly defined panel sections, helping users perceive them as connected components of the same analytical task, while consistency was maintained through repeated visual encodings and interaction patterns. For example, selected SA2s are highlighted in red across all views, creating a clear visual association between visualisations and supporting users' mental models of the dashboard..

Colour choices were informed by both perceptual considerations and data characteristics. The choropleth map uses a sequential blue palette to encode the annual population change for each SA2 region. The sequential colour scheme allows users to quickly distinguish regions experiencing high population growth. The use of a consistent colour scale across all years also improves comparability.

Visual encoding choices were guided by principles of graphical perception. Position along a common scale was prioritised in the scatterplot and ranked dot plot because viewers perceive positional differences more accurately than differences in area or colour intensity (Cleveland & McGill, 1984). The ranked dot plot uses aligned position rather than bars, reducing unnecessary visual ink while maintaining precise comparisons between service intensity values. The choropleth map was selected because geographic location is a crucial component of the research question and spatial patterns would be difficult to communicate using non-spatial charts alone.

Typography and styling were influenced by Tufte's principle of maximising the data-ink ratio and minimising non-data ink (Tufte, 2001). Decorative elements were intentionally reduced, panel structures were kept simple, and visual hierarchy was achieved through layout, spacing, and typography rather than excessive graphical embellishment. This approach helps maintain focus on the data while improving readability and accessibility.

The final dashboard also reflected Shneiderman's (1996) visual information-seeking mantra of "overview first, zoom and filter, then details-on-demand". Users begin with a metropolitan overview through the map and scatterplot, with the option to narrow their focus using SA4

filtering and SA2 selection, and finally access detailed information through the time-series chart, ranked dot plot, and detail panel. This interaction structure supports both exploratory analysis and targeted investigation.

Overall, the final design represents a synthesis of ideas developed across all five design sheets. The dashboard preserves the geographic context introduced in the scrollytelling approach, incorporates the analytical depth of the drill-down design, and retains the comparative capabilities of the interactive slideshow concept. By combining these elements within a coordinated and interactive dashboard, the final design provides a flexible, user-centred narrative visualisation that effectively answers the research question by communicating the relationship between population growth and public transport service provision across metropolitan Victoria.

Changes Following DVP Part 1 Feedback:

Several revisions were made following feedback received in DVP Part 1. The original design sheets contained fewer alternative concepts, with lack of narrative flow and limited justification for why particular visualisations were selected or rejected. In response, the design process was expanded to include a broader range of narrative structures and visualisation combinations, with more explicit discussion of the strengths and limitations of each approach.

One major criticism was the lack of a clearly defined narrative. The initial designs focused primarily on individual visualisations rather than telling a story with fully developed narratives. To address this, subsequent design sheets explored distinct narrative approaches including scrollytelling, drill-down story, and interactive slideshow structures. This allowed the design process to evaluate how different narrative forms supported the communication of relationships between population change and public transport provision.

Feedback also suggested that individual design sheets relied too heavily on single or few visualisations. For example, the original scrollytelling concept consisted solely of a choropleth map with a transport network overlay. This was expanded into a combination of a choropleth map and heat map to better support both spatial pattern recognition and regional comparison. Similarly, later design sheets incorporated multiple coordinated visualisations to provide complementary analytical perspectives rather than relying on a single chart.

Section 3: Implementation

3.1 Technical Implementation

The final dashboard was implemented in R using the Shiny framework to support interactive exploration of population change and public transport service provision across metropolitan Victoria. Several R packages were used throughout the implementation. The tidyverse package supported data manipulation and transformation, sf was used for handling spatial data and geographic boundaries, Leaflet provided the interactive choropleth map, and Plotly enabled the creation of interactive scatterplots and time-series visualisations. Together, these libraries allowed the dashboard to support linked interactions, filtering, and dynamic visual updates.

A notable refinement between the final design and the implemented design is the choice of colour palette used in the choropleth map. In the initial implementation, a sequential colour

scale was considered for representing population change. However, this was later revised to a diverging Red-Blue palette in the final implementation to more appropriately reflect the nature of the variable, which contains both positive and negative values. This change improves the interpretability by visually centering the data around a meaningful midpoint (zero change), allowing users to more easily distinguish between areas experiencing growth versus decline.

Additional data preparation was required beyond the original Data Exploration Project wrangling. The original datasets were transformed into several specialised datasets to support different visualisation tasks. Spatial data was converted to a common coordinate reference system and simplified into map-ready formats for efficient rendering within Shiny. Separate datasets were also created for temporal analysis, service intensity rankings, and map visualisation to improve performance and reduce repeated calculations during user interaction.

Several derived metrics were created to support the narrative objectives of the dashboard. These included annual population change and service intensity rankings. Additional summary statistics, such as the metropolitan median service intensity, were calculated to provide contextual benchmarks within the detail panel and ranked dot plot and support interpretation by non-technical users.

The project presented several implementation challenges. The most significant challenge involved coordinating linked interactions across multiple visualisations built using different libraries. User selections made on either the Leaflet map or Plotly scatterplot needed to update the time-series chart, ranked dot plot, and detail panel simultaneously through Shiny's reactive framework. Additional challenges arose from ensuring acceptable performance when rendering interactive maps and plots, and synchronising filtering behaviour across all dashboard components.

Official documentation for the R packages, alongside examples from package documentation and course materials were used to inform implementation details.

3.2 Interactive Narrative Visualisation Implementation

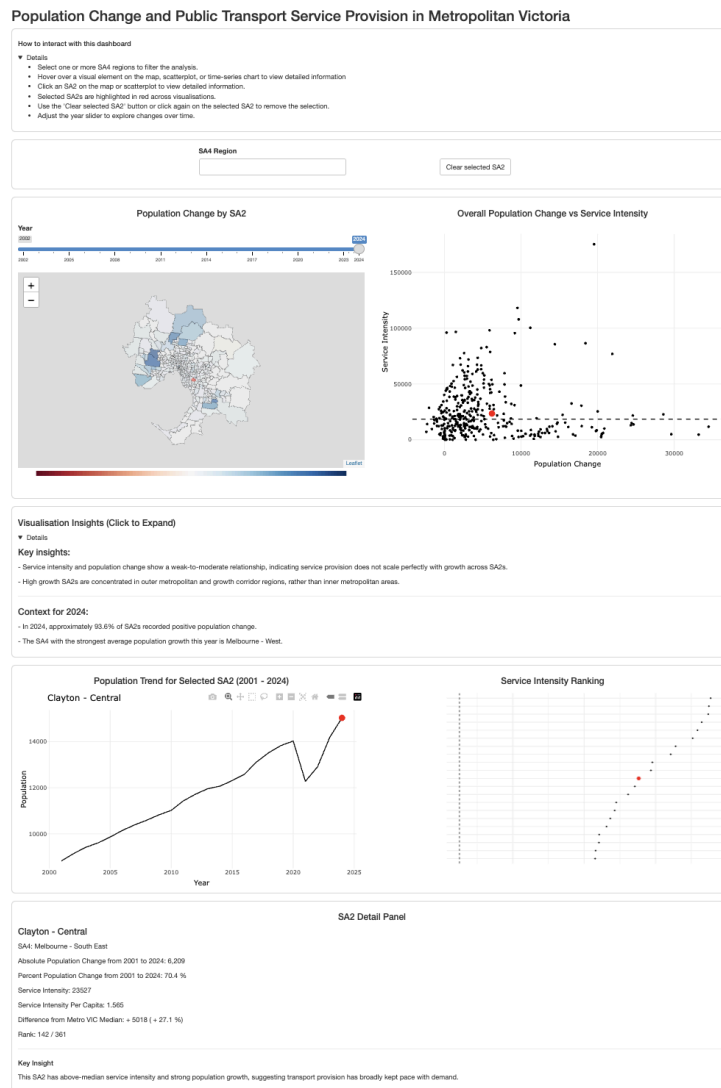


Figure 1: Overall interactive dashboard layout

The final implementation (as seen in Figure 1) was developed as an interactive dashboard with a drill-down narrative structure which allows users to explore the relationship between population change and public transport service provision across SA2 regions in metropolitan Victoria. The dashboard combines multiple coordinated visualisations and interaction techniques to support both guided interpretation and user-driven exploration. The dashboard follows a drill-down narrative structure that guides users from metropolitan-scale overview to regional filtering, and detailed exploration of individual SA2 regions. The narrative is designed around the main research question of whether public transport services are keeping pace with population growth in metropolitan Victoria.

Users begin with a metropolitan-wide overview through the choropleth map and scatterplot. These visualisations communicate the broad spatial and statistical patterns in the data. Users can then narrow down the scope of the analysis through SA4 filtering and investigate individual SA2 regions through linked interactions. Finally, detailed contextual information is provided through the time-series and ranking visualisations and detail panel.

This structure follows Shneiderman's (1996) information-seeking mantra of "overview first, zoom and filter, then details-on-demand," providing a clear analytical workflow while maintaining flexibility for exploration.

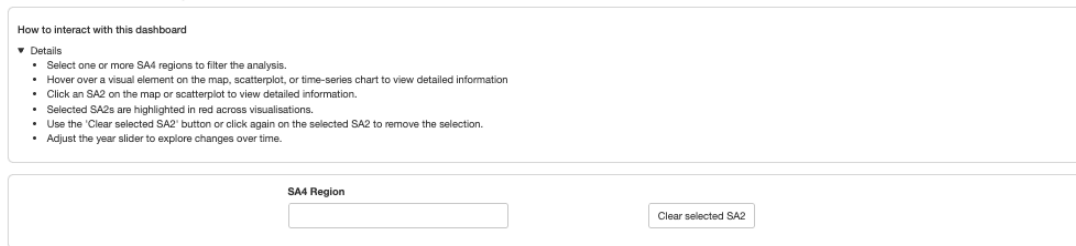


Figure 2: User guidance panel with SA4 filter button and clear SA2 selection button panel

At the top of the dashboard, users are provided with interaction guidance explaining how the dashboard can be used (shown in Figure 2). A SA4 region filter allows users to focus on specific metropolitan regions, reducing visual clutter and supporting regional comparisons. A clear selection button becomes available whenever an SA2 has been selected, allowing users to easily reset the state of the dashboard. These controls support users with varying levels of expertise and help make the dashboard accessible to a general audience.

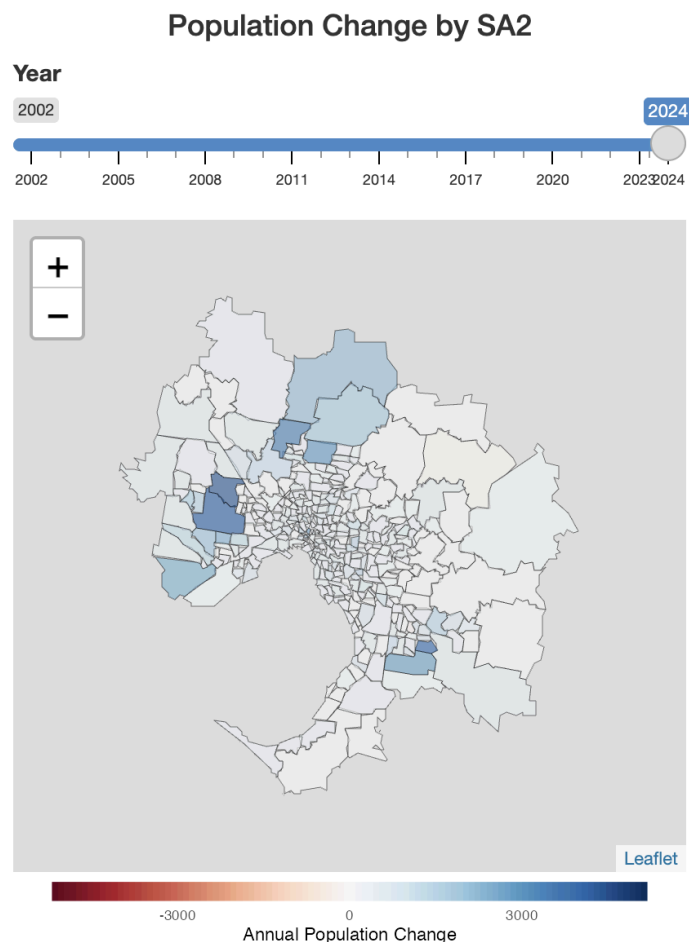


Figure 3: Choropleth map

The map shown in Figure 3 provides the primary spatial overview of the data. A diverging red-blue colour scale is used to distinguish between areas experiencing population decline and areas experiencing population growth. Darker colours indicate larger changes, enabling users to quickly identify growth corridors and areas of decline.

The year slider allows users to explore changes over time, revealing how patterns of population change shift across metropolitan Victoria from 2001 to 2024. Users can also click on any SA2 region to select it, triggering linked updates throughout the dashboard. An interactive tooltip appears on hover providing additional information display about the region when required. The tooltip is dynamically linked to the year slider and SA4 filter, ensuring that the displayed values update in response to user input, maintaining consistency across the visualisation and reinforcing temporal context during interaction. The map supports the narrative by establishing geographic context and highlighting where population growth is occurring.

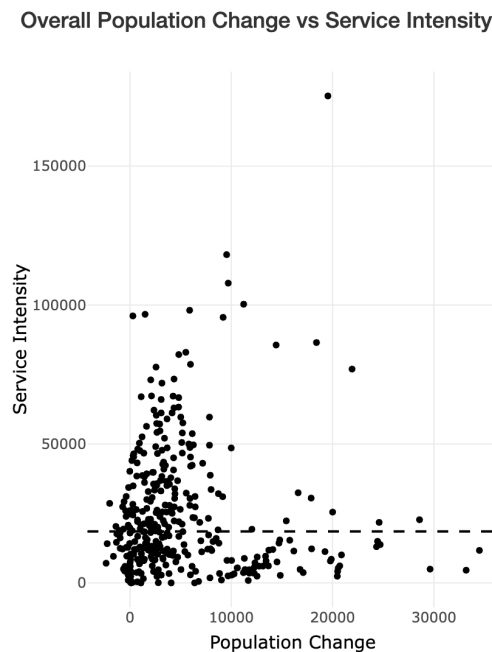


Figure 4: Scatter plot

This scatterplot visualisation seen in Figure 4 allows users to investigate the relationship between transport service provision and population change. Each point represents a SA2 region. The horizontal axis represents population change, while the vertical axis represents service intensity.

The scatterplot enables users to identify potential mismatches between growth and service provision. For example, regions with high population growth but relatively low service intensity may indicate areas where transport services have not developed proportionally with demand. Interactive tooltips provide additional information when users hover over points, while clicking a point selects the corresponding SA2 and updates the linked visualisations. The scatterplot acts as the primary analytical view within the dashboard by directly addressing the main research question.

Population Trend for Selected SA2 (2001 - 2024)

No SA2 selected
Select an SA2 from the map or scatterplot to view population trends.

Population Trend for Selected SA2 (2001 - 2024)

Clayton - Central

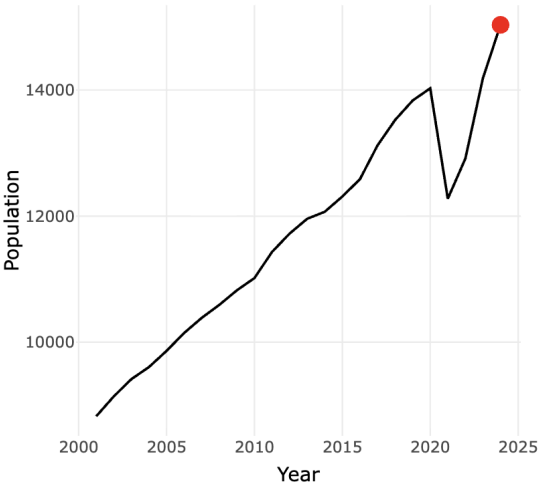


Figure 5a: Time-series chart before SA2 selection

Figure 5b: Time-series chart after SA2 selection

The time-series visualisation provides temporal context by conveying how population has changed over time within the selected region. While the choropleth map focuses on annual changes, the time-series chart allows users to observe long-term population trends as seen in Figure 5b. This additional level of detail helps users understand whether current growth patterns represent recent changes or sustained long-term trends. The selected year is highlighted within the time-series chart, linking the temporal overview provided by the map to longer-term population trends and reinforcing consistency between views. Interactive tooltips are also available, allowing users to inspect exact yearly population values without overwhelming the view. If no SA2 is selected, a message prompts users to make a selection before the chart is displayed (Figure 5a).

Service Intensity Ranking

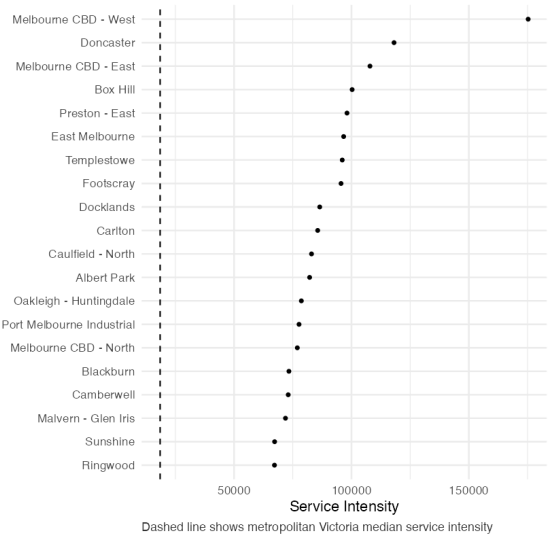


Figure 6a: Ranked dot plot before SA2 selection

Service Intensity Ranking

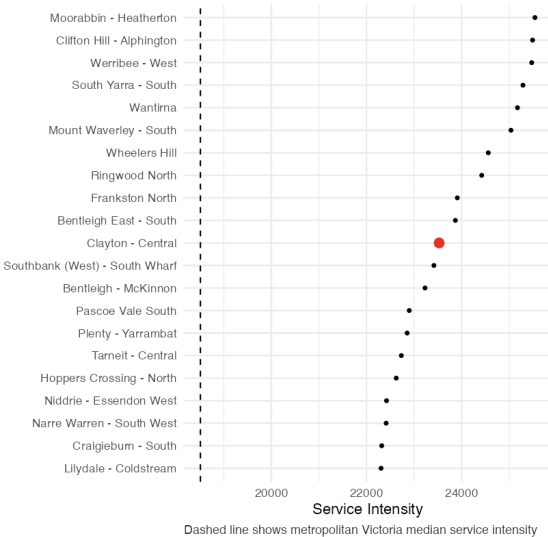


Figure 6b: Ranked dot plot after SA2 selection

The ranked dot plot provides a benchmarking view by positioning metropolitan regions relative to one another. A dashed vertical reference line represents the metropolitan median service intensity, allowing users to quickly determine whether a region is above or below the broader metropolitan benchmark. The metropolitan benchmark for service intensity is defined using the median rather than the mean to provide a more robust point of comparison.

across SA2 regions. This decision was made because the distribution of service intensity is likely to be skewed, with a small number of SA2s potentially exhibiting unusually high values due to major transport hubs or densely serviced urban centres. In such cases, the mean would be sensitive to these extreme values and could misrepresent the “typical” level of service provision across metropolitan Victoria.

To improve readability and reduce visual clutter, the ranked dot plot is restricted to display only the top 20 SA2 regions by service intensity when no specific SA2 is selected as shown in Figure 6a. This decision was made to support effective visual comparison without overwhelming the users with excessive data points, as the full dataset contains a large number of SA2s. Limiting the display to the top 20 ensures that the most relevant regions remain visible and interpretable while still preserving meaningful variation in service intensity. This aligns with principles of visual perception, where reducing cognitive load improves the ability to detect patterns and outliers. In dense ranked visualisations, too many points can lead to overplotting and make relative comparisons difficult, particularly when labels are not directly shown.

When an SA2 is selected, the plot automatically re-centres the view around the selected region (see Figure 6b), making it easier to understand its relative position without displaying all regions simultaneously. This maintains context for comparison while still preserving focus on the selected SA2. This visualisation supports comparison tasks and provides important context for interpreting service provision levels.

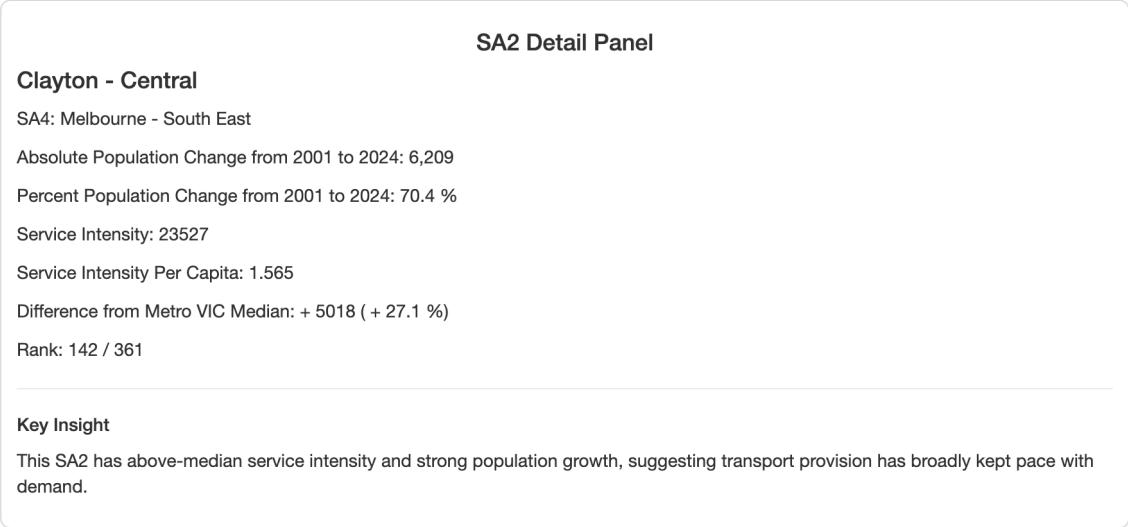


Figure 7: Detail panel

The detail panel shown in Figure 7 summarises key metrics for the selected SA2 region, including SA4 region, population change, service intensity, service intensity per capita, metropolitan ranking, and deviation from the metropolitan median service intensity.

A key insight statement is automatically generated based on the selected region’s characteristics. These narrative statements translate quantitative measures into plain language explanations, offering interpretations of findings for non-technical audiences. For example, regions experiencing strong population growth but below-median service intensity are highlighted as areas where transport provision may not be keeping pace with demand. This component strengthens the narrative aspect of the dashboard by transforming

analytical results into accessible insights. If no SA2 is selected, a message prompts users to make a selection before the chart is displayed.

Visualisation Insights (Click to Expand)

▼ Details

Key insights:

- Service intensity and population change show a weak-to-moderate relationship, indicating service provision does not scale perfectly with growth across SA2s.
- High growth SA2s are concentrated in outer metropolitan and growth corridor regions, rather than inner metropolitan areas.

Context for 2024:

- In 2024, approximately 93.6% of SA2s recorded positive population change.
- The SA4 with the strongest average population growth this year is Melbourne - West.

Figure 8: Insights panel

The dashboard also includes an expandable insights panel (see Figure 8) that provides summary findings and contextual information for the selected year. These insights highlight broad metropolitan patterns and help orient users before they begin exploring individual regions. By combining automated summaries with interactive exploration, the dashboard balances author-driven storytelling with reader-driven analysis.

A key feature of the final implementation is the use of coordinated multiple views. Selecting an SA2 from either the map or scatterplot updates all linked visualisations simultaneously. The selected region is highlighted consistently across the dashboard using a red colour encoding, helping users maintain context while exploring different views. This coordinated interaction design reduces cognitive effort and supports efficient exploration by eliminating the need for users to repeatedly identify the same region across multiple visualisations.

Overall, the implemented dashboard integrates spatial, temporal, comparative, and contextual perspectives into a unified narrative visualisation. The combination of linked interactions, filtering, and explanatory insights enables users to investigate whether public transport service provision is keeping pace with population growth while maintaining a clear and accessible analytical workflow for both technical and non-technical audiences.

3.3 Using the implementation

The interactive narrative visualisation is implemented using Shiny and can be launched by running the application file in RStudio. Once launched, the dashboard opens in a web browser window and presents an overview of population change and public transport service provision across metropolitan Victoria.

An expandable “How to interact with this dashboard” panel is provided at the top of the interface with brief instructions on interactions to assist new users.

Users can begin by selecting one or more SA4 regions using the filter located at the top of the dashboard or adjust the year slider positioned above the choropleth map. These controls update all visualisations simultaneously, allowing analysis to focus on specific metropolitan regions and time periods.

The choropleth map provides a spatial overview of annual population change by SA2. Regions experiencing population growth and decline are represented with a diverging colour

scale. Users can click any SA2 region on the map to select it for further investigation. The map also supports interactive navigation through zooming and panning, allowing users to freely explore spatial patterns at both metropolitan and regional scales. Users can adjust the zoom level either by scrolling or by using the built-in zoom controls, and they can pan across regions by clicking and dragging the map. This enables flexible movement between a broad overview of population change across metropolitan Victoria and more detailed inspection of individual SA2 regions.

The scatterplot allows users to examine the relationship between population change and service intensity. Individual SA2 regions can also be selected directly from the scatterplot. Selected SA2s are highlighted in red and the selection is propagated across all linked visualisations. Users can also expand the insights panel to view automatically generated summary findings and contextual information for the selected year.

Once a SA2 is selected, additional views become available. The time-series chart displays population trends for the selected region from 2001 to 2024, while the ranked dot plot shows the region's service intensity ranking relative to other metropolitan SA2s. The detail panel provides supplementary information including population growth metrics, service intensity measures, ranking information, comparison with the metropolitan median, and an automatically generated interpretation of the region's transport provision.

Several interaction features may not be immediately obvious to first-time users. Clicking a selected SA2 region again will deselect it, and a dedicated "Clear Selected SA2" button is also provided when a selection is active. All visualisations are linked, meaning selections made in either the map or scatterplot automatically update the time-series chart, ranked dot plot, and detail panel.

Additional information can be accessed through interactive tooltips available on the choropleth map, scatterplot, and time-series chart. Hovering over visual elements reveals detailed values such as annual population change, service intensity, and yearly population statistics. These tooltips provide details-on-demand without cluttering the visualisations with excessive labelling.

Section 4: Conclusion and Reflection

This project developed an interactive narrative visualisation that explores the relationship between population change and public transport service provision across metropolitan Victoria. The final dashboard combines spatial, statistical, temporal, and contextual views to support both storytelling and exploration. Through linked interactions between the choropleth map, scatterplot, time-series chart, ranked dot plot, and detail panel, users can investigate whether areas that experienced historically stronger population growth are receiving comparable levels of public transport service provision.

The visualisation successfully communicates several key findings. Population growth is unevenly distributed across metropolitan Victoria with specific growth corridors rather than a uniform spread across the metropolitan area. While there is some relationship between population growth and service intensity, the association is not strong, suggesting that public transport provision does not always increase proportionally with population growth. The dashboard also enables users to identify individual SA2 regions where population growth

and service provision appear to be weakly aligned, highlighting areas where service provision may not have kept pace with population growth. The relationship observed in the scatterplot may also be influenced by a small number of SA2 regions containing major transport hubs, which exhibit disproportionately high service intensity. These areas act as potential outliers and skew the overall distribution, suggesting that service provision is not only solely driven by population growth but also by other factors such as infrastructure concentration and network planning priorities.

One of the most valuable lessons from this project was the importance of the iterative design process. Developing multiple design sheets encouraged exploration of different narrative structures before implementation. The process highlighted the trade-offs between author-driven storytelling and reader-driven exploration. Early scrollytelling and slideshow concepts provided strong narrative guidance but limited user control, whereas the final coordinated multi-view dashboard provided greater flexibility while still supporting a coherent narrative. It provided valuable experience in developing coordinated multiple view dashboards within Shiny, especially with implementing linked interactions between visualisations through reactive programming. The project also reinforced the importance of grounding design decisions in visualisation theory, including Munzner's What-Why-How framework, Gestalt principles, Tufte's design recommendations, and interaction design concepts.

In hindsight, several aspects could be improved. Additional user testing would have helped to identify usability issues and may have informed further refinement of the layout and interaction design. The dashboard currently focuses on metropolitan Victoria as a whole, future work could incorporate comparisons across different years simultaneously, more advanced analytical summaries, or additional transport-related variables to provide richer context. Further enhancements could also include more sophisticated narrative guidance, such as dynamic annotations that automatically highlight notable patterns, emerging trends, or potential outliers as users interact with the visualisations.

A key direction for future work could be the incorporation of socioeconomic status. While the current visualisation focuses on the relationship between population change and public transport service provision, it does not explicitly account for underlying socioeconomic differences across SA2 regions captured in the second research question from the Data Exploration Project. Including a measure such as SEIFA (Socio-Economic Indexes for Areas) would enable a more insightful understanding of whether transport provision is equitably distributed across regions with different socioeconomic statuses. This would also allow the analysis to move beyond purely demographic and service-based comparisons toward an examination of equity and accessibility, which is highly relevant for transport planning policy.

Overall, the project demonstrates how interactive narrative visualisation can be used to communicate complex relationships between demographic population change and public transport provision. By combining multiple coordinated views with interactive exploration, the final implementation provides an accessible and informative tool for understanding transport service provision across metropolitan Victoria.

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Use of Generative AI Statement

I acknowledge the use of generative AI tools to aid in reinforcing concepts and I take full responsibility for the content and its reliability. The following prompts were used and the material was cross checked with reliable online sources as well.

- Is a ranked dot plot better than a ranked bar chart for comparing service intensity between SA2 regions?
- Why does my Plotly visualisation not render with a theme_minimal() theme?
- How do I wrap visualisations in panels in Shiny?
- Explain the Gestalt principles.
- Why is the setView zoom so different between 8 and 9? How can I get a smaller zoom increment?
- How do I centre a header?